Teaching notes and solutions

> **Teacher note · Teacher notes.** Test 2 comes before S19 and S20: assess in it the definitions and the operations already met, without requiring monotone convergence, Bernoulli or the squeeze theorem. Draw horizontal bands around the limit and have students look for an index from which all the terms stay inside them. Require convergence before solving the fixed-point equation. Show $(-1)^n$ to separate "bounded" from "convergent".

Solutions to the activity

1. $u_0 = 5$, $u_9 = 2.3$ and $u_{99} = 2.03$. Since $\frac{3}{n+1} > 0$ at every index, $u_n > 2$: no term equals 2.
2. $|u_n - 2| < 0.01 \iff \frac{3}{n+1} < 0.01 \iff n+1 > 300 \iff n \geq 300$. The first suitable index is therefore $N = 300$. At index 299, the gap is exactly 0.01 and the strict inequality is not satisfied.
3. For $\varepsilon > 0$, take $N = \lceil \frac{3}{\varepsilon} \rceil$, the smallest integer greater than or equal to $\frac{3}{\varepsilon}$. If $n \geq N$, then $n+1 > \frac{3}{\varepsilon}$, hence $|u_n - 2| < \varepsilon$. This choice works, even when it is not minimal.
4. $|v_n - 2| = \frac{1}{n+1} \to 0$, hence $v_n \to 2$. But $v_0 = 3$, $v_1 = \frac{3}{2}$ and $v_2 = \frac{7}{3}$: we observe a decrease then an increase. This convergent sequence is not monotonic.

E1. For $n \geq 1$, $\frac{4n-1}{n+2} = \frac{4-\frac{1}{n}}{1+\frac{2}{n}} \to 4$, since the denominator tends to $1 \neq 0$. Similarly $n^2 - 7n = n^2(1 - \frac{7}{n})$; the first factor tends to $+\infty$ and the second to $1 > 0$, so the product tends to $+\infty$.

E2. The conjugate gives $\sqrt{n+1} - \sqrt{n} = \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. The denominator tends to $+\infty$, so the difference tends to 0.

E3. For $n \geq 1$, $-1 \leq \cos n \leq 1$ and $n > 0$, so $-\frac{1}{n} \leq u_n \leq \frac{1}{n}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.

E4. The sequence is increasing and bounded above by 5, so it converges. By passing to the limit in $u_n \leq 5$, its limit satisfies $\ell \leq 5$. It is not necessarily equal to 5: $u_n = 2 - \frac{1}{n+1}$ is increasing, bounded above by 5 and converges to 2.

E5. Since $|-0.6| < 1$, $(-0.6)^n \to 0$, so $u_n \to 3$. However $u_0 = 1$, $u_1 = 4.2$ and $u_2 = 2.28$: the sequence rises then falls and is not monotonic.

Solutions to Homework 2

1. Let $P(n)$ be the statement "u_n is defined and $0 \leq u_n \leq 2$". It is true at index 0 since $u_0 = 0$. Fix $n \geq 0$ and assume $P(n)$ is true. We have $2 \leq 2 + u_n \leq 4$, so $u_{n+1} = \sqrt{2+u_n}$ is defined and $\sqrt{2} \leq u_{n+1} \leq 2$. Thus $P(n+1)$ is true. By induction, all the terms belong to $[0,2]$.
2. Let $R(n)$ be the statement "$u_{n+1} \geq u_n$". For $n = 0$, $u_1 = \sqrt{2} \geq 0 = u_0$. Fix $n \geq 0$ and assume $R(n)$ is true. The fact that $x \mapsto \sqrt{2+x}$ is increasing on $[0,2]$ gives

$$u_{n+2} = \sqrt{2+u_{n+1}} \geq \sqrt{2+u_n} = u_{n+1}.$$

   So $R(n+1)$ is true. By induction, (u_n) is increasing. It is bounded above by 2, so it converges to a real number $\ell \in [0,2]$.

3. The function $x \mapsto \sqrt{2+x}$ is continuous at ℓ. By passing to the limit, $\ell = \sqrt{2+\ell}$. Since $\ell \geq 0$, this is equivalent to $\ell^2 - \ell - 2 = 0$, that is $(\ell - 2)(\ell + 1) = 0$. The root -1 is excluded; hence $\ell = 2$.

4. Using the conjugate and $u_n \leq 2$,

$$0 \leq 2 - u_{n+1} = \frac{2 - u_n}{2 + \sqrt{2 + u_n}} \leq \frac{2 - u_n}{2}.$$

Let $T(n)$ be the statement "$0 \leq 2 - u_n \leq \frac{2}{2^n}$". At index 0, $2 - u_0 = 2$. If $T(n)$ is true for a fixed integer $n \geq 0$, then

$$0 \leq 2 - u_{n+1} \leq \frac{1}{2}\frac{2}{2^n} = \frac{2}{2^{n+1}}.$$

The base case and the inductive step prove $T(n)$ at every index. Since $\frac{2}{2^{11}} = \frac{1}{1024} < 10^{-3}$, $n = 11$ suffices. This is an index guaranteed by an upper bound, with no claim of minimality. This question can be done before question 3: the number 2 already appears in the bounds.

Solutions to the textbook or past-paper work

Exercise 17. The quotients are at first indeterminate forms. For $n \geq 1$,

$$u_n = \frac{2n - 4}{n^2 + 1} = \frac{2\left(1 - \frac{2}{n}\right)}{n\left(1 + \frac{1}{n^2}\right)} \longrightarrow 0.$$

Similarly,

$$v_n = \frac{2 - 5n}{4n + 7} = \frac{\frac{2}{n} - 5}{4 + \frac{7}{n}} \longrightarrow -\frac{5}{4}, \qquad w_n = n\frac{-1 - \frac{10}{n^2} + \frac{4}{n^3}}{2 + \frac{3}{n} + \frac{1}{n^2}} \longrightarrow -\infty.$$

The last quotient tends to $-\frac{1}{2}$, and the factor n to $+\infty$.

Exercise 24. For every $n \geq 1$, $-1 \leq (-1)^n \leq 1$. Since $n^2 > 0$, we obtain $-\frac{1}{n^2} \leq u_n \leq \frac{1}{n^2}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.